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DNS Log Analysis

Splunk Basics – DNS Log Analysis Lab Report

STEP-BY-STEP GUIDE REPORT

OBJECTIVE

The objective of this lab is to understand how **DNS logs can be analyzed using Splunk** to identify network activity patterns.

During this lab we will:

- Ingest **Zeek DNS logs (JSON format)** into Splunk.
- Learn how Splunk indexes and parses log data.
- Use **SPL (Search Processing Language)** to analyze DNS activity.
- Identify:
    - Most queried domain names
    - Most active source IPs
    - Types of DNS queries in the network

This helps security analysts detect **suspicious DNS activity, malware communication, or abnormal traffic patterns**.

TOOLS AND ENVIRONMENT

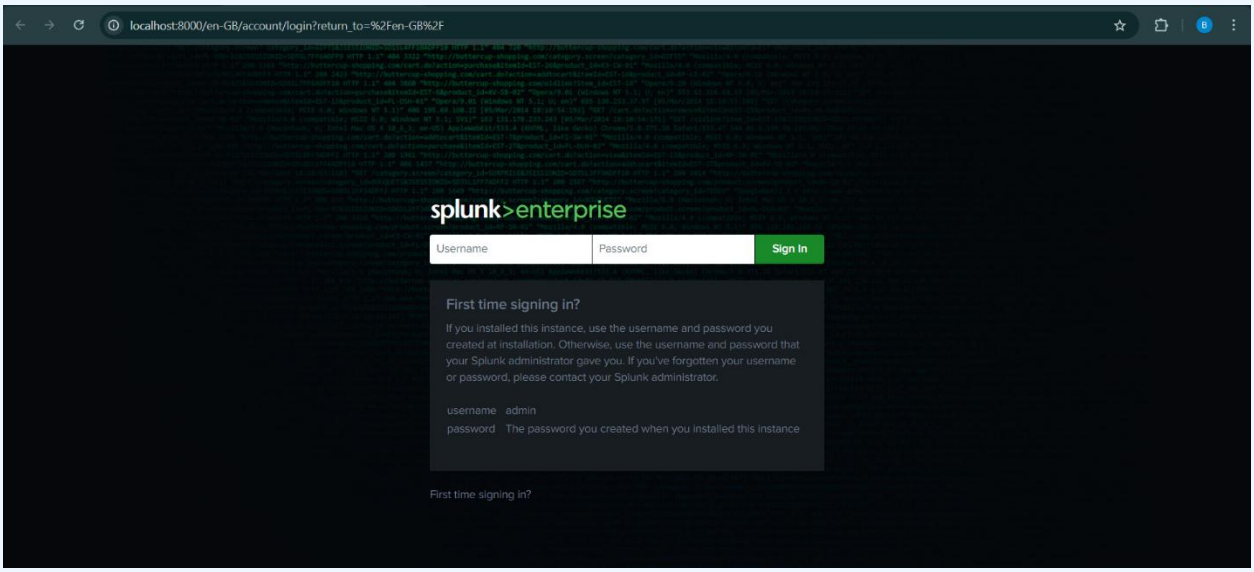
Component	Description
Splunk	Log management and analysis platform
Data Source	Zeek DNS logs
Log Format	JSON
Index Used	dns_lab
Source type	_json

LAB SETUP

Before starting the analysis, the DNS log file must be uploaded to Splunk.

Step 1 – Open Splunk Web

1. Open the **Splunk Web Interface** in your browser.
2. Log in using your credentials.

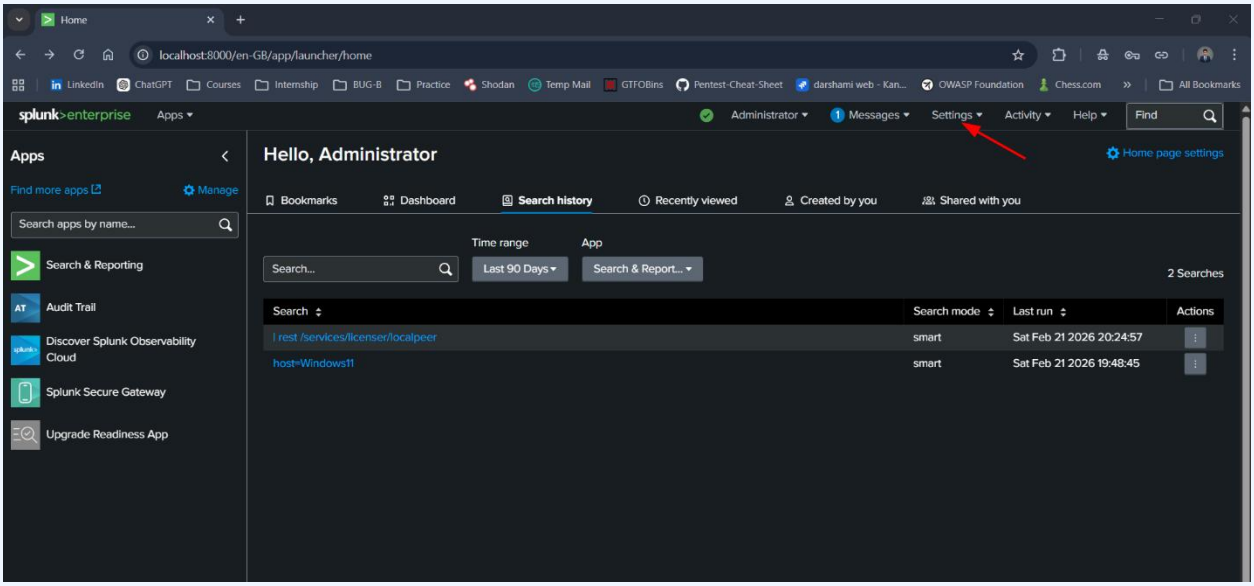


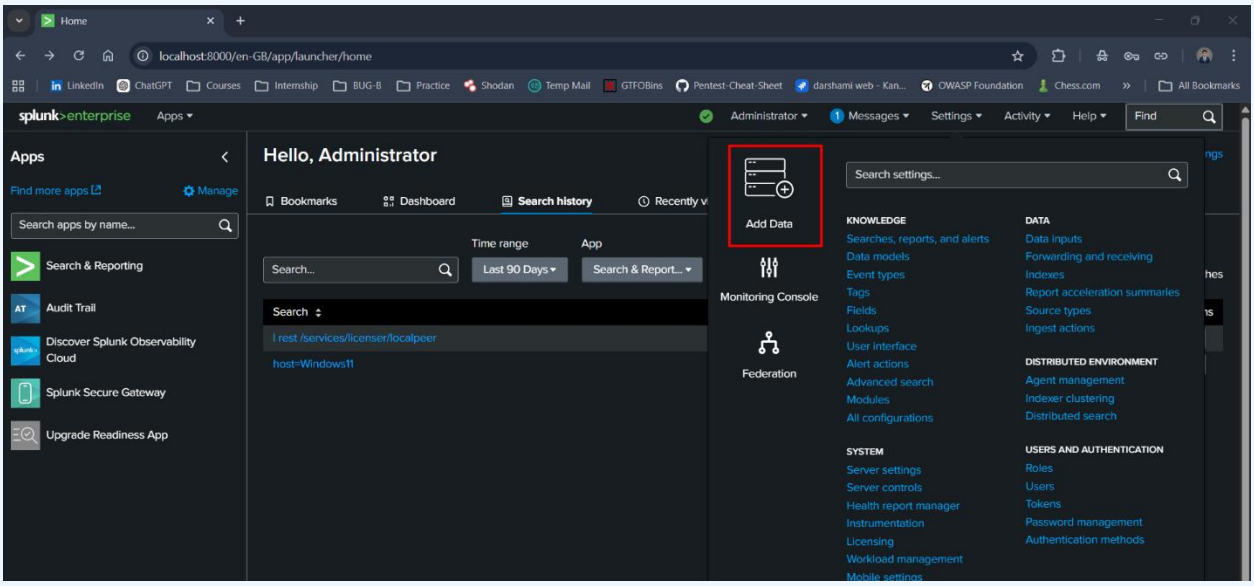
Step 2 – Add Data

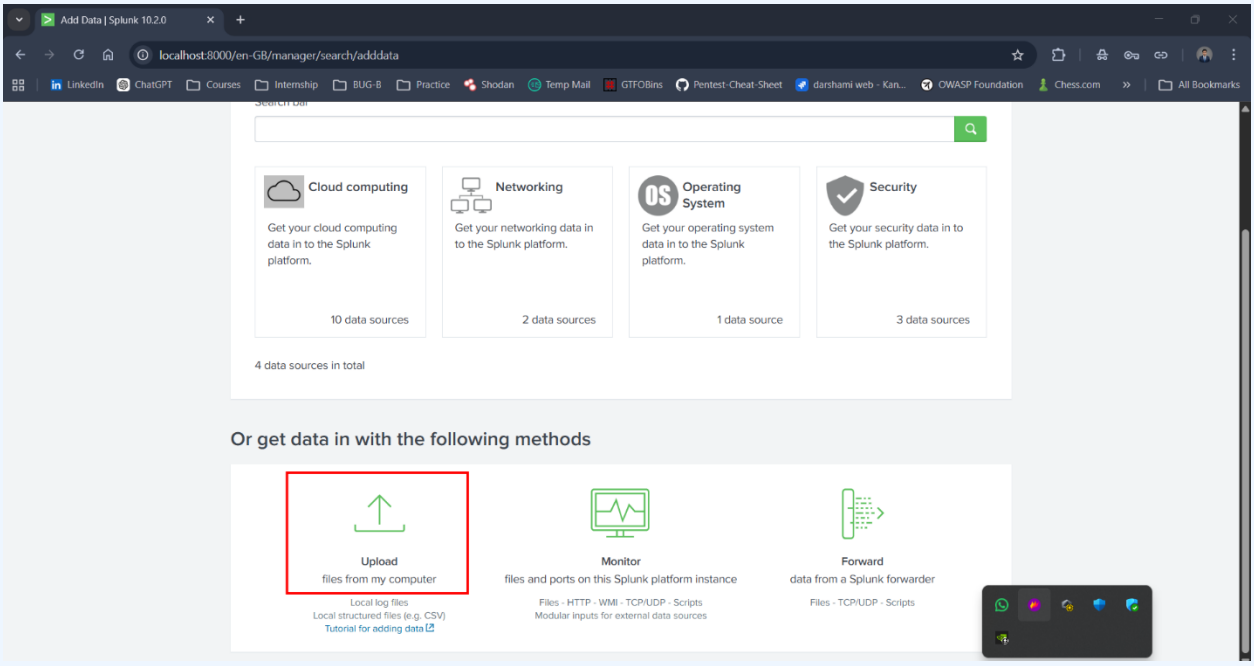
1. Navigate to:

Settings → Add Data

2. Select **Upload**.





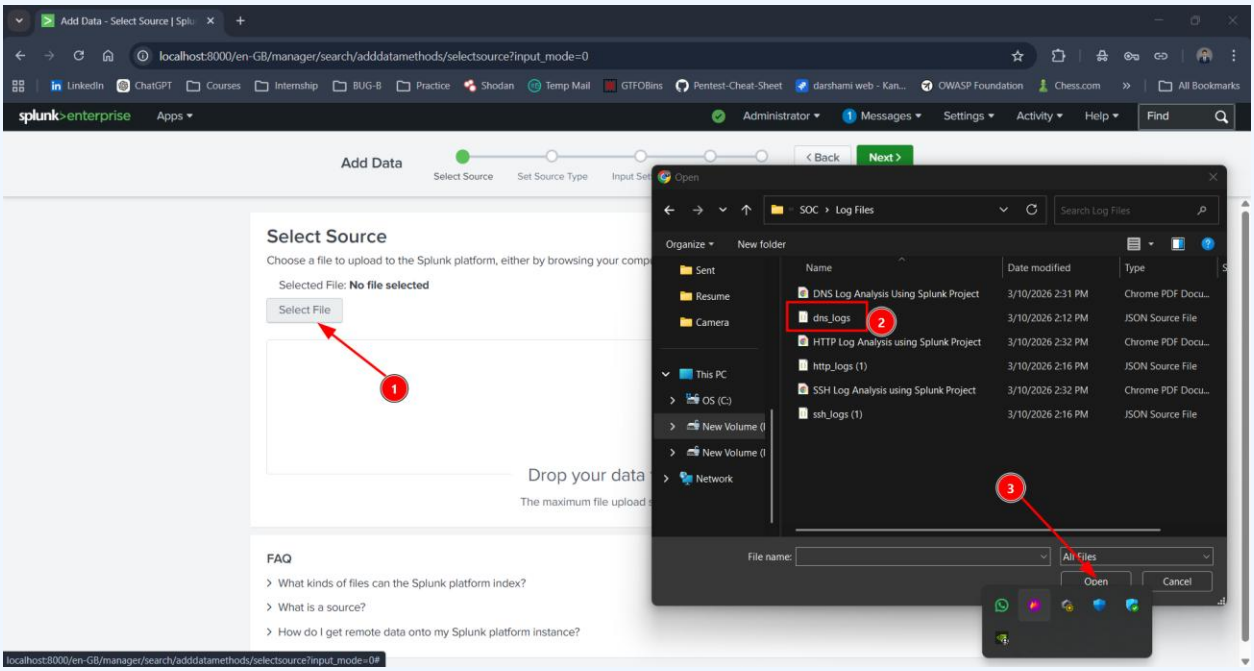


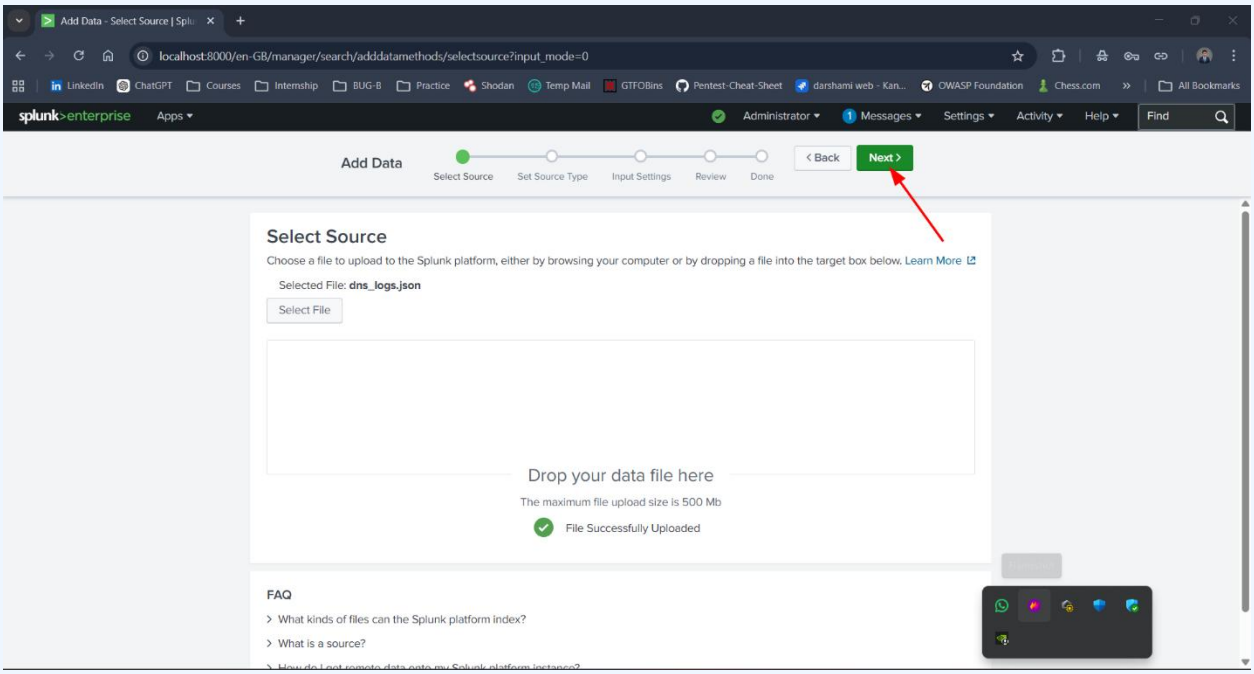
Step 3 – Upload DNS Log File

1. Click **Select File**.
2. Upload the file:

```
dns.log
```

This file contains **DNS query records generated by Zeek network monitoring tool**.


Add Data - Select Source | Splu
localhost:8000/en-GB/manager/search/adddatamethods/selectsource?input_mode=0
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Back
Next
Select Source
Choose a file to upload to the Splunk platform, either by browsing your comp
Selected File: No file selected
Select File
1
Drop your data
FAQ
What kinds of files can the Splunk platform index?
What is a source?
How do I get remote data onto my Splunk platform instance?
Open
SOC › Log Files
Organize
New folder
Sent
Resume
Camera
This PC
OS (C:)
Network
Name
Date modified
Type
DNS Log Analysis Using Splunk Project
dns_logs
HTTP Log Analysis using Splunk Project
http_logs (1)
SSH Log Analysis using Splunk Project
ssh_logs (1)
2
3
File name:
All files
Open
Cancel


Add Data - Select Source | Splu
localhost:8000/en-GB/manager/search/adddatamethods/selectsource?input_mode=0
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Next
Select Source
Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. Learn More
Selected File: dns_logs.json
Select File
Drop your data file here
The maximum file upload size is 500 Mb
File Successfully Uploaded
FAQ
What kinds of files can the Splunk platform index?
What is a source?

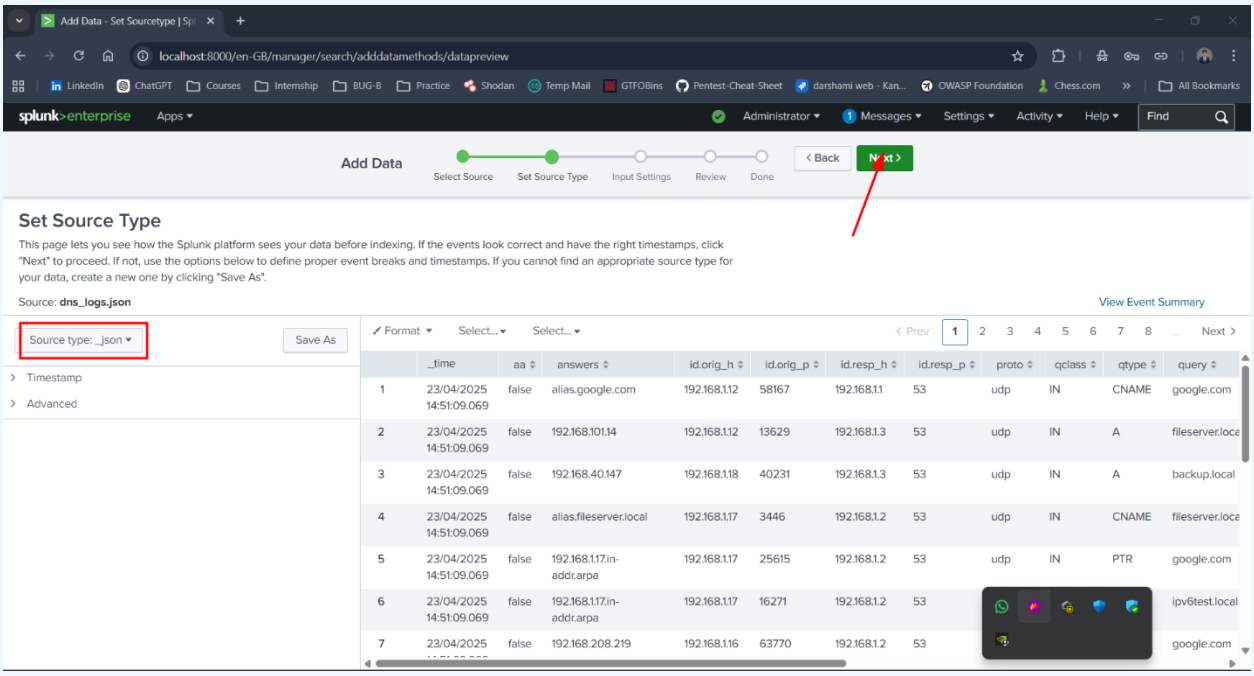
Step 4 – Configure Source Type

Set the **Source Type** as:

```
_json
```

Reason:

- Zeek logs are structured in **JSON format**
- Splunk automatically extracts fields like:
  - query
  - id.orig_h
  - qtype
  - answers



Step 5 – Configure Index

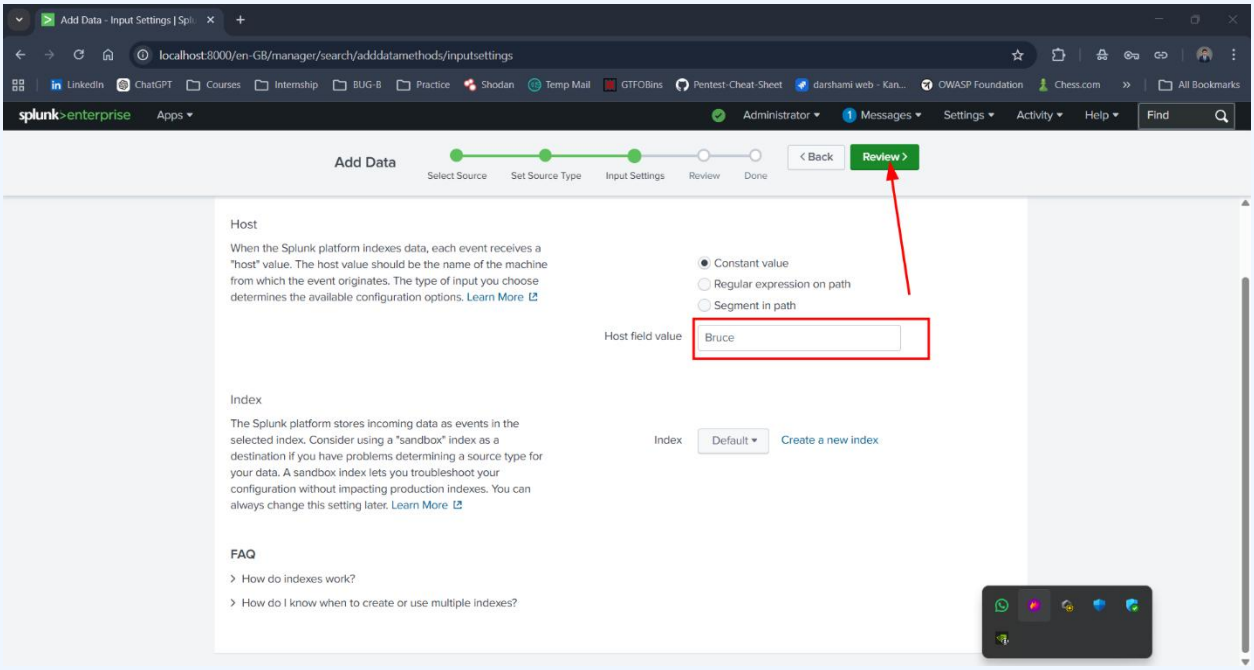
Choose or create an index.

Example:

```
dns_lab
```

Reason:

- Keeps lab data separated from other logs.
- Makes searching easier.



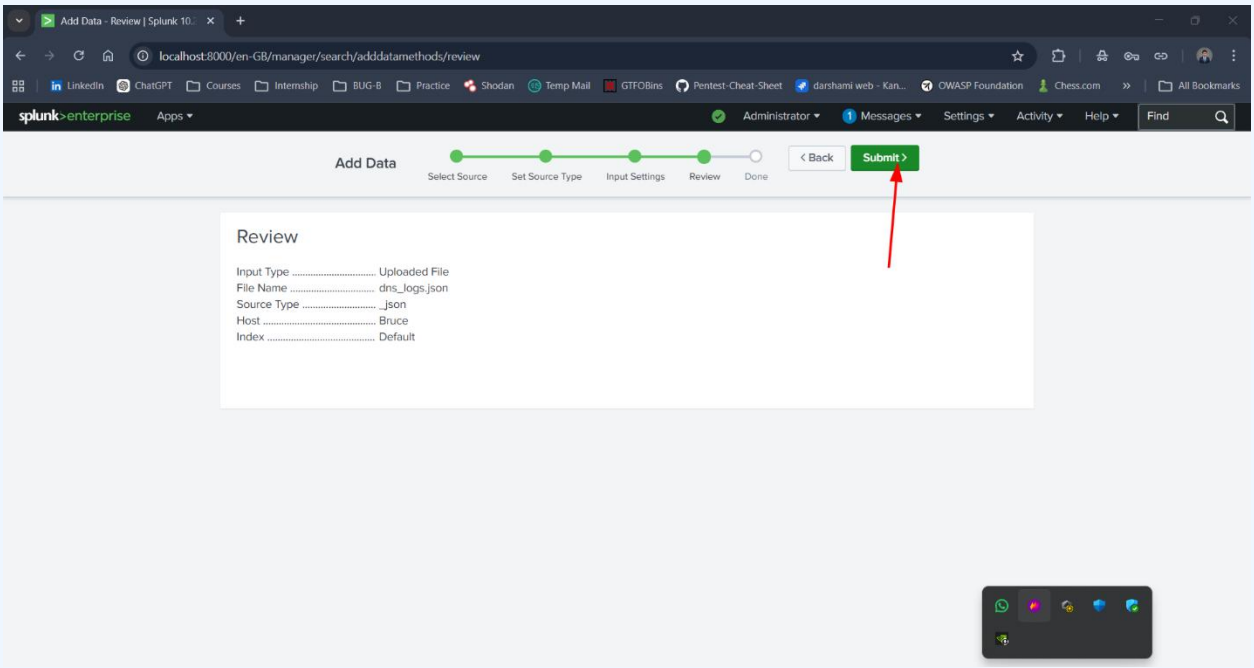
Step 6 – Review and Submit

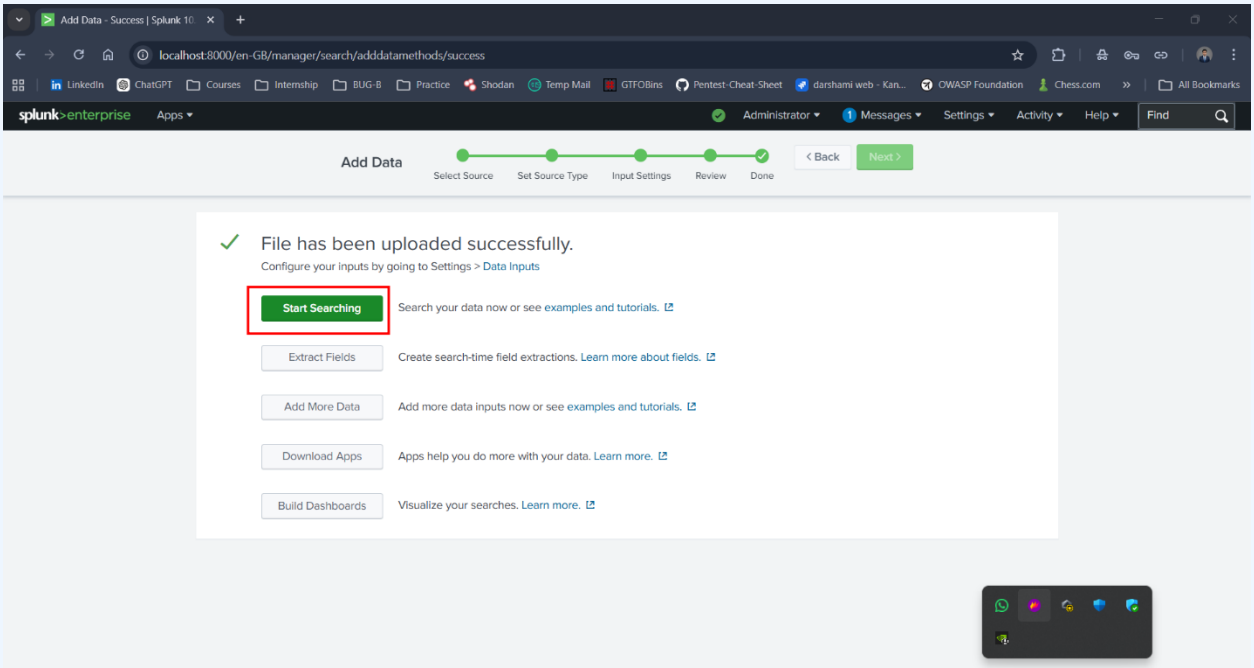
1. Review configuration.
2. Click **Submit**.
3. Splunk will index the data.

You can confirm ingestion by running:

```
index=dns_lab
```

If events appear, the upload was successful.





UNDERSTANDING DNS LOGS

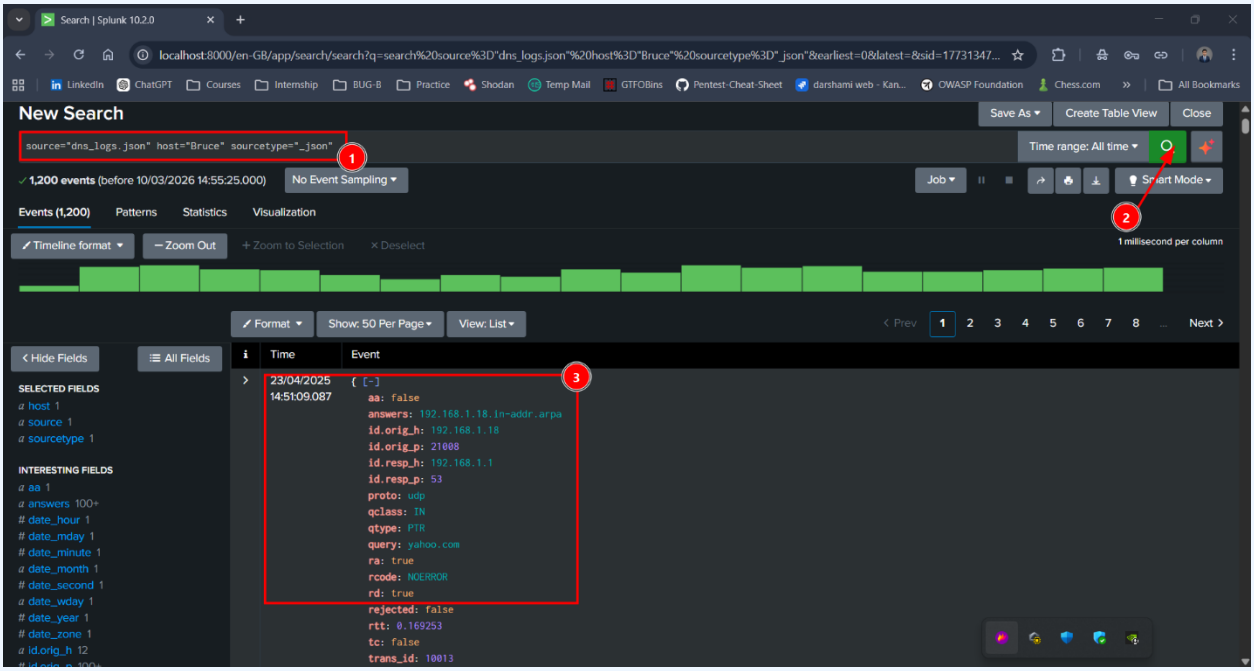
DNS logs record **domain name resolution requests** from clients.

Example DNS Event Fields:

Field	Description
id.orig_h	Source IP address making the DNS request
query	Domain name requested
qtype	Type of DNS query
answers	IP returned by DNS

Example Log:

```
{
"id.orig_h":"192.168.1.10",
"query":"google.com",
"qtype":"A",
"answers":"142.250.190.78"
}
```



LAB TASKS

Task 1: Identify the Most Frequently Queried Domain Names

SPL Query:

```
index=dns_lab sourcetype="json"
| stats count by query
| sort -count
```

Explanation

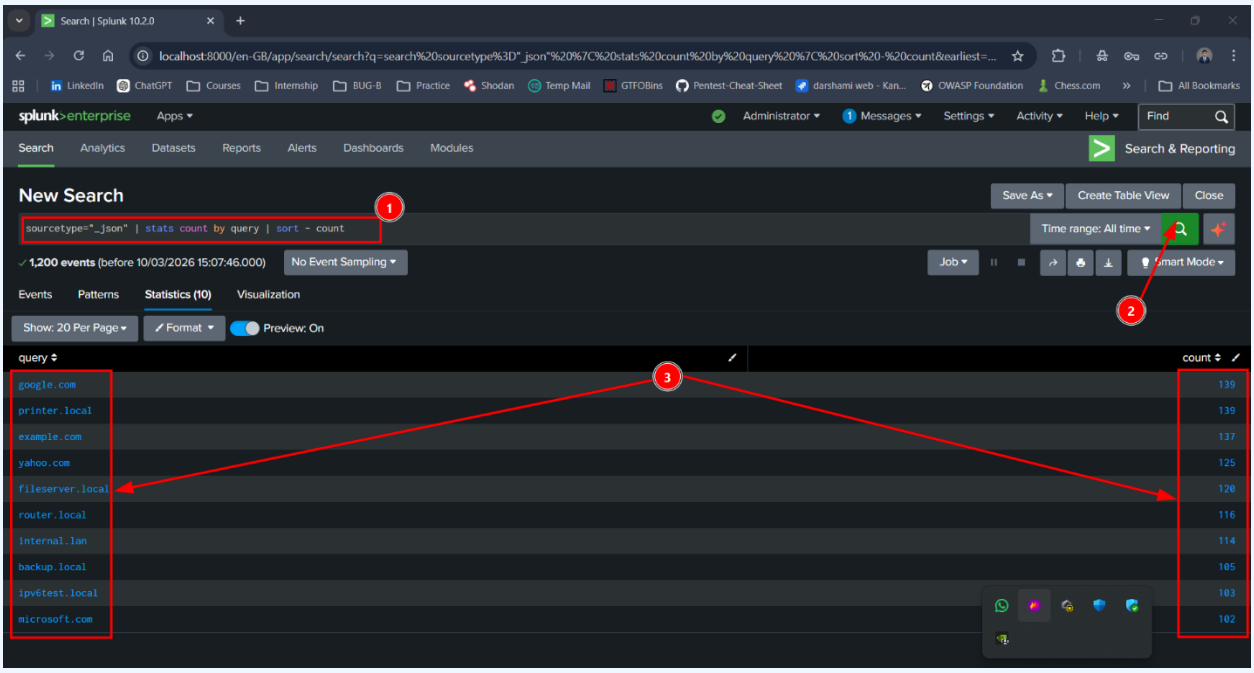
Command	Purpose
index=dns_lab	Searches logs stored in dns_lab index
sourcetype="json"	Filters only JSON formatted logs
stats count by query	Counts how many times each domain was queried
sort -count	Sorts results in descending order

What This Shows

This query reveals:

- Most visited websites
- Potential malware domains

- Unusual DNS traffic patterns



Task 2: Find the Most Active User IPs Generating DNS Traffic

SPL Query

```
index=dns_lab sourcetype="json"
| stats count by "id.orig_h"
| sort -count
```

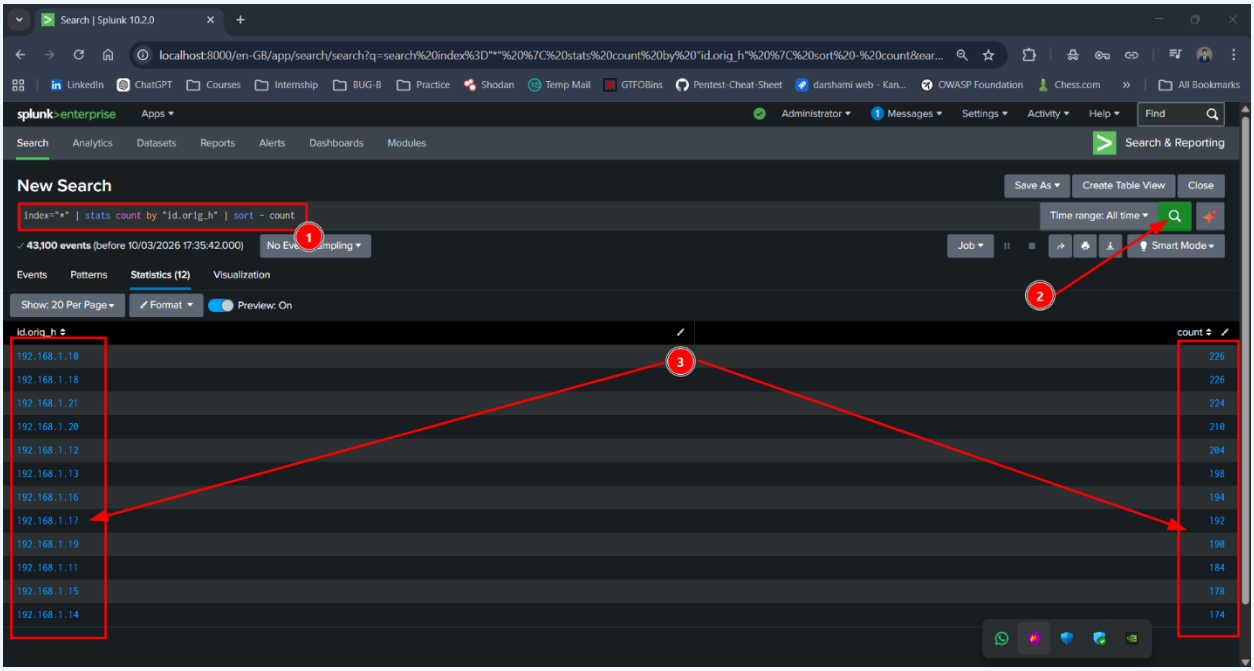
Explanation

Command	Purpose
stats count by "id.orig_h"	Counts DNS queries from each source IP
sort -count	Shows most active IPs first

What This Shows

This helps identify:

- Devices generating high DNS traffic
- Potential infected machines
- Misconfigured systems



If one host generates **extremely high DNS queries**, it may indicate **malware beaconing**.

Task 3: Breakdown of DNS Query Types

DNS queries can request different types of records.

SPL Query

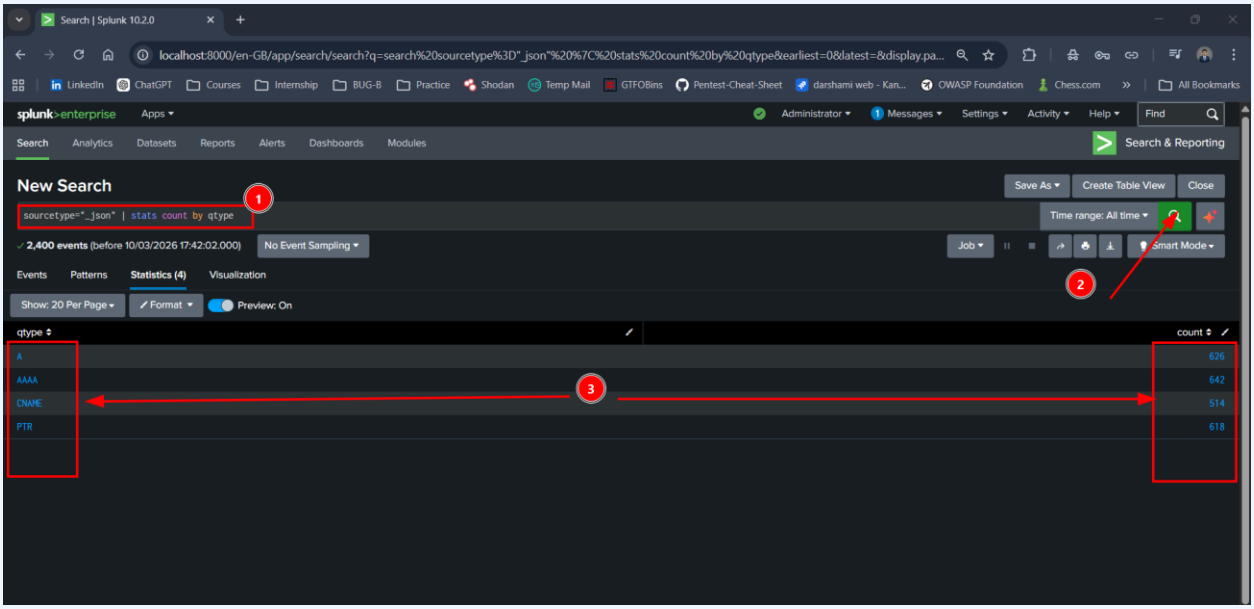
```
index=dns_lab sourcetype="json"
| stats count by qtype
```

Explanation

Command	Purpose
stats count by qtype	Groups DNS events by query type

Common DNS Query Types

Query Type	Meaning
A	Resolves domain to IPv4 address
AAAA	Resolves domain to IPv6 address
CNAME	Alias record
PTR	Reverse DNS lookup



SECURITY INSIGHT

DNS analysis helps detect:

Suspicious Activities

- Malware contacting command and control servers
- DNS tunneling
- Domain generation algorithms (DGA)
- Data exfiltration

Example Indicators:

- Very long domain names
- High number of DNS queries from one host
- Queries to unknown domains

Splunk makes it easier to **visualize and detect such anomalies**.

CONCLUSION

In this lab we successfully:

- Uploaded **Zeek DNS logs into Splunk**
- Learned how Splunk parses **JSON structured data**
- Used **SPL queries** to analyze DNS traffic

The analysis helped identify:

- Most frequently queried domains
- Most active source IPs
- Distribution of DNS query types

This type of analysis is essential for **network monitoring, threat detection, and incident response** in cybersecurity environments.